

CS Imaging Integration Guide

Programmer's guide to integrate CS Imaging 8 with
your Software

1. CHANGE HISTORY

Date	CSi8 version	Suite version	Change
07/2018	8.0.0	2.0.0	CS Imaging version 8 is now supported
07/2020	8.0.7	3.0.15	CS Imaging Client/Server version is now supported
01/2021	8.0.10	3.1.2	Patient folder argument optional, recommended to use patient ID instead
07/2021	8.0.12	3.2.1	Patient folder for a given patient id can be retrieved by calling CSi8 Server via REST API
01/2022	8.0.17	3.4.0	Social Security Number is now supported For new integration, Patient ID integration is the sole integration, no more Patient folder
03/2022	8.0.18	3.5.0	Update Guide

2. INTRODUCTION

This document explains how to integrate CS Imaging with your software (e.g. a Dental Practice Management Software or any other software), to launch CS Imaging or to retrieve information for a given patient.

Carestream Dental is regularly improving integration capabilities to deliver an as-streamlined-as-possible experience to end-users. Carestream Dental is committed to review your suggestions, please send us your comments and feature request to CSImagingIntegrationSupport@csdental.com.

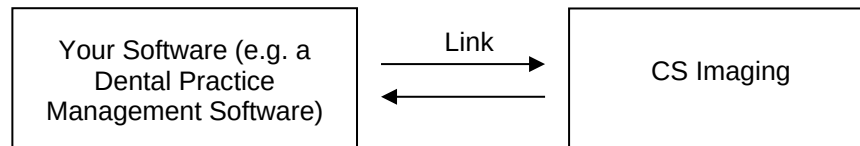
NOTE: For historical integration via patient folder ("-P")

You may have already integrated your software with CS Imaging by using "-P" (patient folder) option in the command line. Carestream Dental is not requesting pro-active migration to the new Patient ID integration. The patient folder integration will not evolve but will be supported by Carestream Dental. If you are currently using this historical mode and willing to transition, please contact us CSImagingIntegrationSupport@csdental.com so we can advise on the migration path.

All stations must follow the same integration mechanism either all patient folder ("-P") or all patient ID ("D"), a mix is not supported and will result in unexpected behavior of CS Imaging

The following document does not detail anymore this historical integration method.

3. GENERAL PRINCIPLES



The principle is that your software manages patient's information. The information will be passed as parameters when your software calls CS Imaging, which will take care of patient images / data storage.

In other words, your software (e.g. a Dental Practice Management Software) will always be the master of patient information (demographics, etc.), while CS Imaging will always be the master of patient images / data storage (folder organization, etc.)

To reconcile patient information and patient images / data between both systems, a unique key for each patient is used: patient identifier, a.k.a. Patient ID. This best practice comes from DICOM / HL7 recommendations, used for years in radiology, hospital, etc., where mix of software / equipment manufacturers is common.

4. BENEFITS OF THE INTEGRATION

- Your software can launch CS Imaging via simple command-line, passing through essential patient demographics
- Your software doesn't have to manage patients' image storage, managed by CS Imaging
- Your software can retrieve back the patient folder via REST API if needed (e.g. to get image thumbnails, store documents, etc.)
- Patients created in this method can also be reviewed when launching CS Imaging directly from Windows desktop (a.k.a. standalone)
- Your software can use CS Imaging as a bridge to store images to PACS and/or retrieve images from PACS, when CS DICOM option is enabled.

5. HOW TO IMPLEMENT THE LINK

5.1 CALLING CS IMAGING FROM YOUR SOFTWARE

To pass necessary information from your software to CS Imaging to acquire images, the command line allows the following parameters:

TW.EXE -D<patient ID> [-N<patient last name>] [-F<patient first name>] [-B<patient date of birth>] [-G<patient gender>] [-S<social security number>]

- **TW.EXE** is the application name; the full path to that application should be provided. This path can be retrieved from Windows registry: key "**HKLM/Software/Classes/Trophy**", value "**InstallDir**".
- **<patient ID>** is the patient's unique identifier in the calling application (max **64** characters). This parameter is **required**. It will be used to populate DICOM Tag (**0010,0020**) in any DICOM image acquired via CS Imaging.
- **<patient last name>** is the patient's last name (max **64** characters). This parameter is optional. It will be used to populate DICOM Tag (**0010,0010**) in any DICOM image acquired via CS Imaging.
- **<patient first name>** is the patient's first name (max **64** characters). This parameter is optional. It will be used to populate DICOM Tag (**0010,0010**) in any DICOM image acquired via CS Imaging.
- **<patient date of birth>** is the patient's date of birth (format is **MMDDYYYY**). This parameter is optional. It will be used to populate DICOM Tag (**0010,0030**) in any DICOM image acquired via CS Imaging.
- **<patient gender>** is the patient's gender (possible values are **M** for male, **F** for female and **O** for unknown). This parameter is optional. It will be used to populate DICOM Tag (**0010,0040**) in any DICOM image acquired via CS Imaging.
- **<social security number>** is the patient's social security number (SSN). This parameter is optional; however it is required by some countries (e.g. Nordic countries).

EXAMPLE

CS Imaging is installed in folder "C:\Program Files (x86)\Carestream\CSImaging8". The patient's name is John Sample. His identifier in the calling application is SJ000. He is male and date of birth is January 25, 1966. His social security number is 100966.

The command line should be:

"C:\Program Files (x86)\Carestream\CSImaging8\TW.exe" -DSJ000 -FJohn -NSample -B01251966 -GM -S100966

NOTE:

- It is best to launch the imaging software from an icon in your software
 - There is a space between each parameter
 - The parameters (**-N**, **-F**, **-B**, **-D**, **-G** and **-S**) must be in capital letters
 - The patient folder will be created by CS Imaging when the patient file is created
 - The patient files are managed by CS Imaging and can be displayed when running the software without DPMS (a.k.a. standalone).
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5.2 RETRIEVING CS IMAGING INFORMATION IN YOUR SOFTWARE (OPTIONAL)

When your software is integrated with CS Imaging, you may need to retrieve folder in which patient data are stored, for example:

- To read DICOM file to present additional info to the end user directly in your software
- To store some documents (such as reports, etc.) in patient folder (see next paragraph)

To query information about a specific patient, our CS Imaging Server is exposing a REST API:

- Request
 - o Method: GET
 - o URL: /patient_dirs/{dicom_patient_id}
 - o Example: http://CSImagingServer:9002/patient_dirs/007
- Response in JSON
 - o Code 200: successful operation. Patient folder shall be returned in “data”. Example below:

```
{
  "code": 0,
  "message": "OK",
  "data": [
    "\\sharestorage\\csi8db\\3e8cf394-c590-4327-86af-70f2221ceb49"
  ]
}
```

- o Code 400: error code, with error message. Example below:

```
{
  "code": 10201,
  "message": "Patient not found!"
}
```

NOTE: You should never modify any file already present in patient folder as we cannot guarantee CS Imaging behavior if an external source is alienating existing patient data

5.3 RETRIEVING PATIENT IMAGE INFORMATION IN YOUR SOFTWARE (EXAMPLES)

5.3.1 Retrieving DICOM file information

CS Imaging produces 2 types of files

1. DICOM files created by Dental devices integrated in CS Imaging, named with “dcm” file extension
2. Other files (xml, etc.) that are private Carestream Dental working file (which can change from one version of our products to another)

Once you've retrieved patient folder, you can read DICOM file content to display them in your software.

NOTE: The image processing in DICOM file may be different than what is displayed in CS Imaging, as processing is updated at runtime.

5.3.2 Retrieving image thumbnails

CS Imaging offers the ability to automatically generate a copy of image in open format (JPEG/TIFF/BMP/PNG) in a subfolder of the given patient (refer to CS Imaging User Guide for the configuration). Once it is configured, you can get the image in open format from the patient folder.

EXAMPLE

CS Imaging is configured to save a copy of images in **JPEG**. The patient folder is retrieved via REST API (see [Retrieving CS Imaging information](#)) and it is “\\sharestorage\\csi8db\\3e8cf394-c590-4327-86af-70f2221ceb49”.

The JPEG images can be found in folder:

“\\sharestorage\\csi8db\\3e8cf394-c590-4327-86af-70f2221ceb49**JPG**”.

5.4 STORING YOUR OWN DATA IN PATIENT FOLDER (OPTIONAL)

Once you've retrieved patient folder, you can directly save in this folder documents such as word, pdf, etc. By doing so, those documents will be displayed on CS Imaging gallery for this patient.